

Capstone Project: Data Engineering

Project Specification & Requirements

1. Project Overview

Each student is required to design, develop, and deploy a unique **End-to-End ETL Pipeline** following the **Medallion Architecture**. You will select a unique business domain (e.g., E-commerce, Fintech, Healthcare, Logistics) and build a robust data architecture using **Apache Airflow** and **Snowflake**.

Your system must be automated, idempotent, self-healing, and provide reporting and analytics.

2. Technical Architecture & Standards

A. Snowflake Environment & Layering

You must implement a three-layer Medallion Architecture within Snowflake to ensure data integrity and traceability.

- **Bronze (RAW):** Ingest raw, immutable dummy data. Use a landing zone approach.
- **Silver (STAGING):** Perform cleaning, schema enforcement, casting, and deduplication.
- **Gold (PROD):** Final Star Schema (Facts and Dimensions) optimized for analytics.
- **Naming Convention:** Database must be named `MBUST_{{student_roll}}`.

B. Data Modeling & SCD Type 2

- **Dimensional Modeling:** Your Gold layer must consist of at least **one Fact table** and **three Dimension tables**.
- **Historical Tracking:** Implement **Slowly Changing Dimensions (SCD Type 2)** for at least one core dimension to track historical changes over time.
- **Calendar:** Use an existing Date/Calendar dimension from the Snowflake Marketplace (building your own is optional).

C. Orchestration & Engineering (Airflow)

- **Modular DAGs:** Break down your pipeline into logical, decoupled tasks (Extract → Load → Stage → Transform).
- **Idempotency:** The pipeline must be re-runnable for the same period without creating duplicate records or corrupting the Star Schema.
- **XComs & Hooks:** Use standard Airflow Providers and manage credentials via Airflow Connections (no hardcoded secrets).

3. Development Standards (Mandatory)

Code Quality & Linting

Treat this repository as a production codebase. All Python code must be strictly **PEP8** compliant.

- **Pre-commit Hooks:** Your repo must contain a `.pre-commit-config.yaml` file.
- **Automation:** You must use `black` (formatting), `flake8` (linting), and `check-yaml` to pass all checks before committing.
- **Docstring:** Use the **Google** or **NumPy** docstring format.

Observability & Reporting

- **Failure Alerts:** Configure the Airflow DAG to send an Email Alert immediately upon any task failure.
- **Success Reporting:** On a successful run, generate a Daily Summary Report (e.g., total rows processed, key business metric) delivered to the stakeholder via Snowflake Alert.

4. Analytics & Documentation

- **Snowsight Dashboard:** Create a professional dashboard in Snowflake using your Gold layer data to visualize KPIs.
- **README.md Requirements:**
 - **System Architecture Diagram:** High-level flow from Airflow to Snowflake.
 - **ER Diagram:** Visual representation of your Star Schema.
 - **Data Dictionary:** Definitions for your Fact and Dimension columns.
 - **Setup Guide:** Step-by-step instructions to reproduce the environment.

5. Submission & Evaluation Criteria

Criteria	Expectation
SCD Type 2	Correct handling of historical records and valid/expired timestamps.
Medallion Integrity	Clear separation of Bronze, Silver, and Gold layers.
Code Quality	PEP8 compliance and functional pre-commit hooks.

Orchestration	Idempotent DAGs, proper use of Hooks, and secret management.
Observability	Working Email alerts and success summary reports.
Data Sharing	Successful Snowflake secure share with the instructor account.

Note: Each student must submit their **Unique Theme**. No two students may use the same business logic and same code.

Final Grading & Evaluation Schema

Assessment Component	Weight	Description
Capstone Project	35%	Implementation of the End-to-End ETL Pipeline (Medallion Architecture, SCD Type 2, Airflow Orchestration, and Snowflake Dashboard).
Viva Voce	15%	Oral examination covering your project logic, technical choices, and core Data Engineering concepts discussed in sessions.
Written Exam	50%	Final comprehensive exam
Total	100%	